

Notes and References.

1. The *Poly Question Society* (abbreviated wherever possible as P?S, or PQS if only a limited character set is available and pronounced as if spelled pee-kew-ess) is an informal group of questioners concerned with a multitude of subjects. In general, the members of the P?S (whether alive, dead or imaginary) met at an institution of higher learning known as the *Polytechnic Institute of Brooklyn* or more informally "Brooklyn Poly" (now a division of *New York University*). Nearly all P?S members are computer programmers known for exceptional works implemented on various computer architectures, especially the DEC PDP-8 family.

The P?S is an outgrowth of a pre-existing organization known as the *William Claude Fields Memorial Programming Group* (WCFMPG) (also known as *Friends of the Boston Massacre*). The various works of the WCFMPG are legendary in the realm of PDP-8 programming; all original WCFMPG members were responsible for either starting or creating various software projects for DEC computers, especially the PDP-8. All members of the WCFMPG were DEC employees at some point in their careers; however, much of their work on PDP-8 systems (and other computers made by other companies) predates their employment at Digital.

2. SIMH stands for "history simulation" (in the sense of the history of computing) for the purpose of preserving as much as possible the functionality of computers generally no longer manufactured. The SIMH project is maintained by a loose affiliation of programmers, most generally specializing in one or two computer architectures out of the many systems supported.

Levels of support for specific computers varies; at one point the SIMH project was very DEC-centric but now includes support for models from various other manufacturers.

More information on the SIMH project can be obtained from various Internet websites hosting components of the overall SIMH project including various source code files and other forms of documentation.

3. SIMH simulator support is available for additional PDP-8 features and devices normally disabled by default; however, additional options are documented within the SIMH environment by invoking certain internal *help* commands. Where relevant, additional documentation is provided in Appendix A above. For those seeking further information, links to certain Internet websites are provided that may supply information regarding seldom used features of SIMH; however, most users of PEPS will not be concerned unless intending to develop software expressly targeting specific devices.

4. OS/8 is the principal marketing name of an operating system that can run on most PDP-8 models (or in some instances a specific variant name is used because of the need to support slightly different and likely incompatible models which cannot run the main system components). The group of name variant systems constitutes a confusing mess due to some of these variations being created merely for the purposes of highly dubious marketing ploys; only an expert can tell the difference between a variant with little or no technical purpose over OS/8 Version 3D (also known as Version 3Q) and another slightly different system that can only be used with newer models such as what are known as Omnibus machines, or perhaps DECmates depending on the specific variations, albeit not necessarily completely functional). In general it is prudent to avoid these newer variants when using the SIMH simulator due to the likelihood of subtle incompatibilities suddenly appearing seemingly without warning.

Certain specific programs acquired from newer release variants as mentioned above provide acceptable functionality when used in OS/8 V3D. As such, the image copy of OS/8 provided in the PEPS package contains several notable programs that correct bugs and quirks in the originally released versions associated with OS/8 Version V3D.

OS/8 was originally created as a quick-and-dirty hack pulled off over a single weekend intended as a demo program to be shown to DEC management the next Monday morning. While there was every intention to write a second system with more reasoned design internals, foolishly, management insisted all further changes be made to the demo version instead of a complete rewrite. As a direct result, many serious problems became institutionalized into the design that cannot be corrected.

The original demo (and some of the follow-on work) was done by Richard Lary, who also created most of the R-L Monitor System (also known later as *MS/8*) along with the rest of the members of the WCFMPG. The R-L Monitor System was the first PDP-8 operating system ever used by most of the members of the P?S.

Part of the reason the deficient demo was created is because the R-L Monitor System was shown to DEC management personnel, all of whom rejected it essentially because they had no idea what it was or what it actually did; the demo was designed to show how a computer user could approach the PDP-8 in a manner vaguely similar to the way this was done on other DEC computers available at the time such as using TOPS10 on the PDP-10 from the point of view of a single user. As such, the managers obtained a completely superficial and flawed understanding of the demo (which for them was likely the most they could muster considering their minimal understanding of the other operating systems mentioned above that were in place at the time).

It is noteworthy that all of the other members of the WCFMPG refused to work on the demo because of its flimsy nature. Later, some other members of the P?S were hired by DEC to work on OS/8 along with a few unrelated staffers of varying abilities. While this mixed group did eventually develop the demo into something more functional, the inherent internal limitations cannot be overcome without a complete rewrite (which had been proposed but was rejected by management).

Using OS/8 today, whether on real-world hardware or within the PEPS environment or other simulators, requires mastering certain unusual skills on the part of the user such as working in a mythical year environment sometime prior to 01-Jan-2002. Unfortunately, due to limitations within the file system, it is not possible to determine the proper year of any particular file because there is an inherent anomaly regarding which group of eight years in the range of 1970-1977, 1978-1985, 1986-1993 or 1994-2001 is the true file origination date reported by the available utilities in an obfuscated manner.

In specific instances, particular users know the proper year group for certain files from personal experience, but inevitably, any collection of files will become commingled with files from one of the other three possible date groups; worse still, program files created today must have mythical dates. New users must keep track of the real-world date offset on a file by file basis. Because of this serious anomaly, coupled with the advantages offered by advanced features of the PEPS package, it is strongly recommended that all users embrace host editing of source files. Within the host environment, more descriptive filenames are possible with accurate file dates as long as the file date can be set after 01-Jan-1980. (Note: This still causes problems identifying files known to have been created from 01-Jan-1970 through 31-Dec-1979 or perhaps earlier.)

Certain intrepid users are known to manually insert pragmatic date modifications in the output files of OS/8 directory listings to maintain fantasy records that may contain more information than OS/8 can actually provide. However, due to Y2K considerations, this practice has an upper limit of 31-Dec-2069.

5. P?S/8 is a replacement system for the R-L Monitor system. It includes thousands of improvements borne of all-new work carried out over the course of decades. Despite all of this, the basic P?S/8 system is quite low overhead (or in some subtle ways actually lower overhead), when compared to the original R-L Monitor System. On many specific hardware configurations, P?S/8 runs palpably faster than OS/8 performing comparable tasks.

Certain specific concepts unique to the R-L Monitor System (and not translatable to OS/8) have either been faithfully implemented in P?S/8, or in some cases, actually improved.

P?S/8 is still being developed. While the P?S/8 release included in the current PEPS package is Version 8Z first released on 23-Feb-1988, various improvements and bug fixes have been applied to the system image included in the PEPS release.

As soon as P?S/8 Version 9A is completed, it will replace the present Version 8Z release and be included in future PEPS packages.

P?S/8 runs on all of the same PDP-8 models that OS/8 (and its variants) runs on; however, the single P?S/8 release covers all models, differing only in terms of specific device handlers and related utilities where applicable. It is noteworthy that a P?S/8 system written to RX01 diskette media can be booted to all suitably configured models from the original "straight" PDP-8 through the DECmate II. By way of comparison, three different OS/8 variants are required to support the same models due to insurmountable incompatibilities inherent in OS/8 internal design (and at least four variants were marketed for this purpose). Note: No OS/8 variant exists to support bootable RX01 diskettes on the DECmate II.

A minor patch must be applied to the supplied P?S/8 release for the TC01/TC08 DECTape controller due to SIMH incompatibility with the real-world hardware; the P?S/8 keyboard monitor includes direct commands to rewind and unload DECTapes. When applied on the real-world hardware, appropriate use of these commands helps achieve noticeable performance improvements; unfortunately, using these commands is known to destabilize the SIMH program.

Documentation is available explaining the minor (reversible) patching procedure needed if moving the system image to an actual DECTape for use on real-world systems. Various programs exist to move DECTape images between physical DECTapes and the SIMH-compatible files used in the PEPS environment. Appendix D provides links to appropriate utilities for this and related purposes.

6. BIN format is the standard binary loading format defined in the paper-tape software system. On occasion, a BIN format file must be loaded into memory in the SIMH simulated environment. Both the OS/8 and P?S/8 operating systems support utilities to process BIN format data passed to the SIMH environment by setting up the *PTRIN.TXT* file with the intended binary contents. When files are moved from the SIMH environment to the host windows system, the *PTPOUT.TXT* file is used.